

17 Introduction to Multi-Parameter models

- Most interesting problems involve multiple unknown parameters.
- When there are two (or more) unknown parameters the prior and posterior distribution will each be a *joint* probability distribution over *pairs* (or tuples/vectors) of possible values of the parameters.

Example 17.1. Assume body temperatures (degrees Fahrenheit) of healthy adults follow a Normal distribution with unknown mean μ and unknown standard deviation σ . Suppose we wish to estimate both μ , the population mean healthy human body temperature, and σ , the population standard deviation of body temperatures. Assume a discrete prior distribution according to which

- μ takes values 97.6, 98.1, 98.6 with prior probability 0.2, 0.3, 0.5, respectively.
 - σ takes values 0.5, 1 with prior probability 0.25, 0.75, respectively.
 - μ and σ are independent.
1. Start to construct the Bayes table. What are the possible values of the parameter (in this discrete example)? What are the prior probabilities? (Hint: the parameter θ is a *pair* (μ, σ) .)
 2. Suppose two temperatures of 97.5 and 97.9 are observed, independently. Identify the likelihood.
 3. Complete the Bayes table and find the posterior distribution after observing these two measurements. Compare to the prior distribution.

4. The prior assumes that μ and σ are independent. Are they independent according to the posterior distribution?

Example 17.2. Continuing Example 17.1, let's assume a more reasonable, continuous prior for (μ, σ) .

1. What are the possible values of σ ? How would you choose a prior for σ ?
2. For a prior, assume that: μ has a Normal distribution with mean 98.6 and standard deviation 0.3, σ has a “Half-Normal(0, 1)” distribution (that's Normal(0, 1) but constrained to positive values), and μ and σ are independent. Simulate (μ, σ) pairs from the prior distribution and plot them.
3. Simulate the prior predictive distribution. Do the values seem reasonable? If not, adjust the prior distribution of parameters and try again.
4. Find and interpret a central 98% prior credible interval for μ .

5. Find and interpret a central 98% prior credible interval for σ .

6. What is the prior credibility that both μ and σ lie within their credible intervals?

Example 17.3. Continuing Example 17.1, we'll now use grid approximation to approximate the posterior distribution.

1. Assume a grid of μ values from 96 to 100 in increments of 0.01, and a grid of σ values from 0.01 to 3 in increments of 0.01. How many possible values of the pair (μ, σ) are there; that is, how many rows are there in the Bayes table?

2. Suppose first that we are given a sample of two measurements of 97.9 and 97.5. Use grid approximation to approximate the joint posterior distribution of (μ, σ) . Simulate values from the joint posterior distribution and plot them. Compare to the prior. Compute the posterior correlation between μ and σ ; are they independent according to the posterior distribution?

3. In a recent study, the sample mean body temperature in a sample of 208 healthy adults was 97.7 degrees F and the sample standard deviation was 1.0 degree F; the sample data is summarized in Section 17.1.5 (it's the same data we used in Section 14.1). Use grid approximation to approximate the joint posterior distribution of (μ, σ) . Simulate values from the joint posterior distribution and plot them. Compare to the prior. Compute the

posterior correlation between μ and σ ; are they independent according to the posterior distribution?

Example 17.4. Continuing Example 17.1, we'll now use simulation to approximate the posterior distribution, given the same sample data as before.

1. Describe how, in principle, you would use a naive (non-MCMC) simulation to approximate the posterior distribution. In practice, what is the problem with such a simulation?
2. Use `brms` to approximate the posterior distribution. Compare to the results of the grid approximation.
3. Find and interpret a central 98% posterior credible interval for μ .
4. Find and interpret a central 98% posterior credible interval for σ .
5. What is the posterior credibility that both μ and σ lie within their credible intervals?

Example 17.5. Continuing the previous example

1. Describe how you could use simulation to approximate the posterior predictive distribution.
2. Run the simulation, summarize the predictive distribution, and find and interpret a 95% posterior predictive interval.
3. Explain how you could use posterior predictive checking to assess the reasonableness of the model in light of the observed data. Does the model seem reasonable?

17.1 Notes

17.1.1 Discrete grid

```
mu = c(97.6, 98.1, 98.6)
sigma = c(0.5, 1)

# theta is all possible (mu, sigma) pairs
theta = expand.grid(mu, sigma)
names(theta) = c("mu", "sigma")

# prior
```